Internal Assessment – 1 (CIA-1)

Department: Mathematics
Course Name: Mathematics-I for CSE/ECE stream
Batch: 2025-26
Semester: I

Date: 08/12/2025
Course Code: 25BEPHY/ELY101
Max. Marks: 50
Duration: 90 Mins

Note: Answer all the questions. Q1&Q2 carries 10 Marks each, Q3- Q6 carry 20 Marks each

Q No	Question	Marks	RBT Level	CO
1.	a. Find the angle between the radius vector and the tangent to the curve $r = a(1 - \cos\theta)$	5	L2, L3	CO1
	b. Find the radius of curvature of the curve $y = a\log\left(\sec\left(\frac{x}{a}\right)\right)$ at any point(x, y).	5		
	OR			
2.	a. Find the total derivative of $z = x\,y^2 + x^2\,y$, where $x = at,\ y = 2at.$	5	L2, L3	CO1
	b. Obtain the Maclaurin's expansion of $sin^{-1}x$ up to the term containing x^3.	5		
3.	a. Find the pedal equation of the following curve $r^n = a^n \cos n\theta.$	7	L2, L3	CO1
	b. For the curve $r = ae^{\theta \cot\alpha}$, prove that $\frac{\rho}{r}$ is constant. Where ρ is the radius of curvature and α is constant.	7		
	c. Show that the curves intersect each other orthogonally $r = a(1 + \sin\theta)$ and $r = a(1 - \sin\theta).$	6		
	OR			
4.	a. Find the angle between the radius vector and the tangent to the curve $\frac{2a}{r} = 1 - \cos\theta$ at $\theta = \frac{2\pi}{3}$	7	L2, L3	CO1
	b. Find the radius of curvature for $x^3 + y^3 = 3axy$ at $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ on it.	7		
	c. Find the angle of intersection of the following pairs of curves $r = a\theta$ and $r = \frac{a}{\theta}$	6		
5.	a. Using Maclaurin's series expand $f(x) = log\,(cosx)$ up to the term containing x^4.	7	L2, L3	CO1
	b. Find the total derivative of $u = x^2 + y^2 - z^2$, where $x = e^t,\ y = e^t \cosh t$ and $z = e^t \sinh t.$	7		
	c. Evaluate : $\lim_{x\to 0}(cotx)^{tanx}$	6		
	OR			
6.	a. If $u = \frac{yz}{x},\ v = \frac{zx}{y},\ w = \frac{xy}{z}$, show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4.$	7	L2, L3	CO1
	b. If $u = f(x - y,\ y - z,\ z - x)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0.$	7		
	c. Evaluate : $\lim_{x\to 0}\left(\frac{a^x + b^x}{2}\right)^{\frac{1}{x}}$	6		

Course outcomes: Students will be able to

CO1: Apply the knowledge of calculus to solve problems related to polar curves.